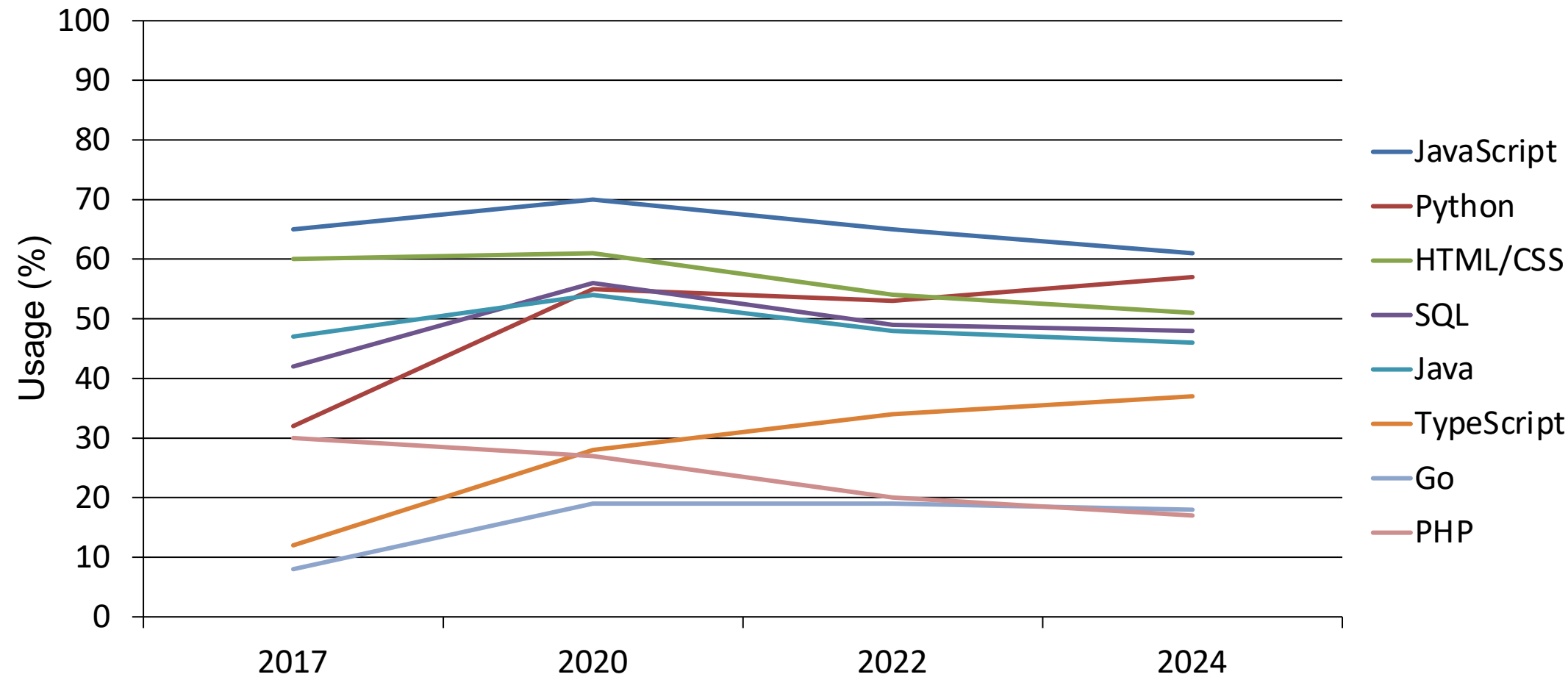


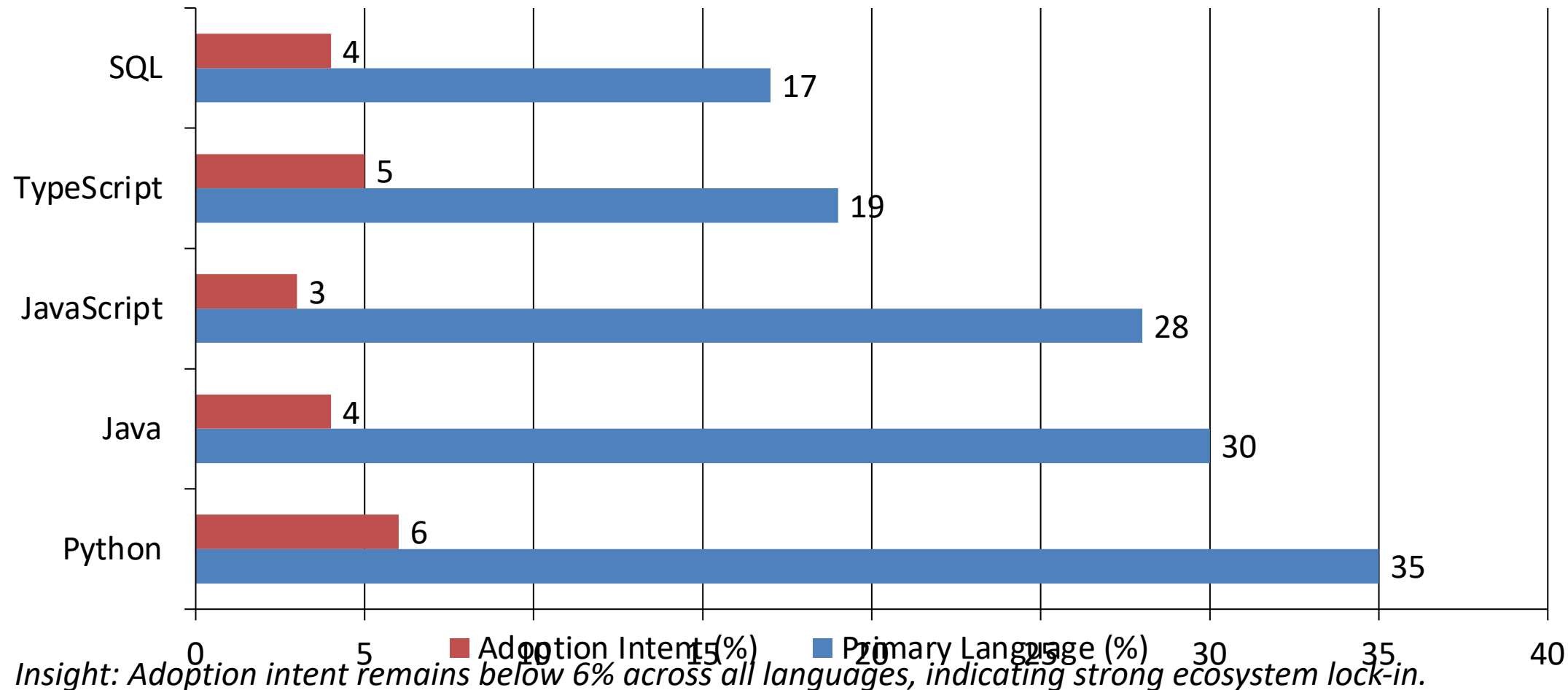
State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

JetBrains' annual survey of 23,262 developers
worldwide

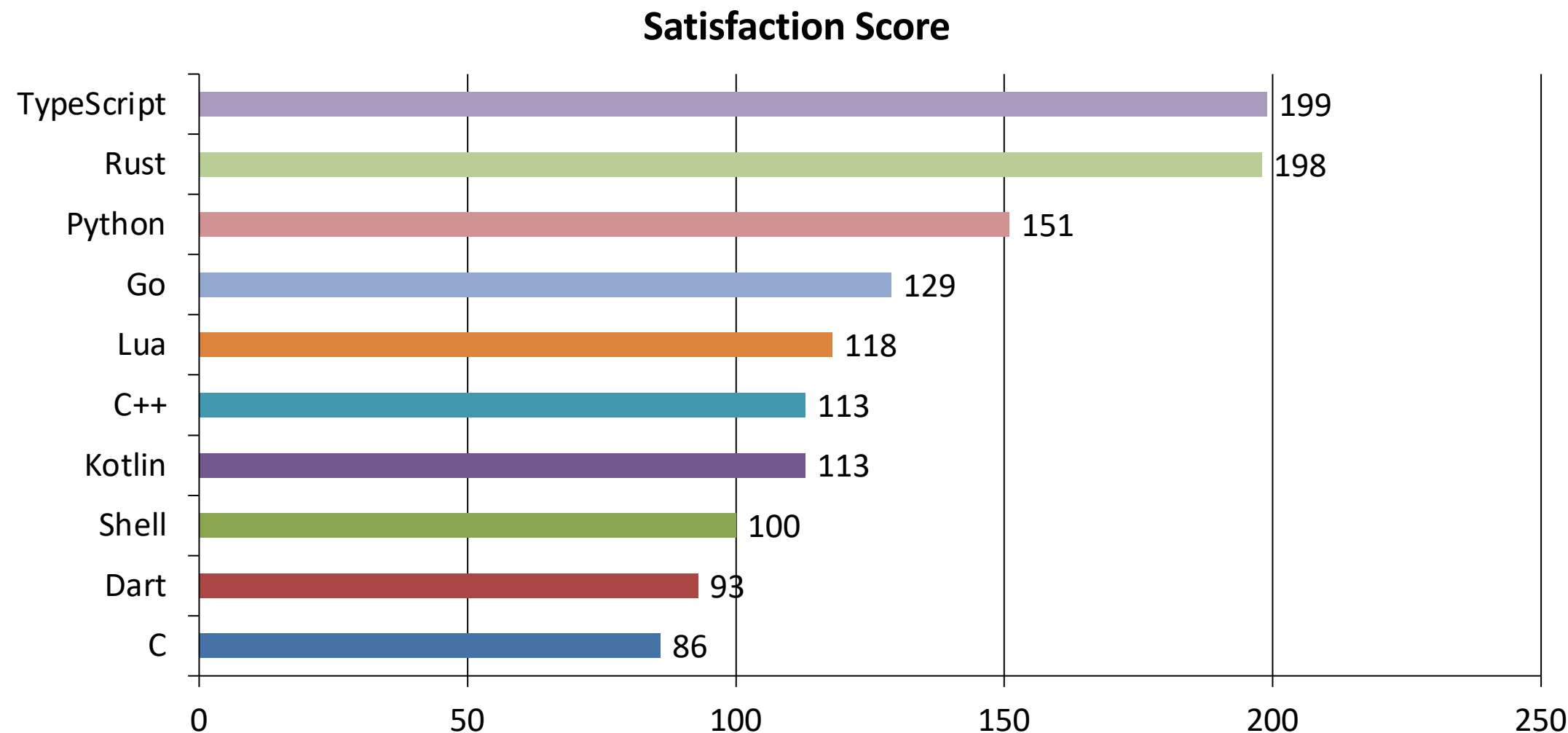
Python Closes the Gap: 8 Years of Language Usage Shifts



Python Is Now the #1 Primary Language at 35%



TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Domain	Top Language	Share (%)
AI / Machine Learning	Python	31%
Data Processing & Analytics	Python	17%
User Interface	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating with APIs & Services	Python	38%

What This Means for Your Technology Strategy in 2025

- Invest in Python for AI/ML teams — it is the undisputed leader.
- TypeScript is the future of frontend and full-stack development.
- Rust and Go are the growth plays for infrastructure and systems.
- PHP and Ruby hiring pools are shrinking.
- Developer satisfaction correlates with type safety and modern tooling — factor this into retention strategy.